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# An Improved YOLOv11 Network for Marine Debris Detection in Underwater Environments

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**Abstract**

Marine debris detection plays a crucial role in environmental protection and intelligent underwater robotics. However, the underwater environment poses unique challenges due to low visibility, object occlusion, and high background noise. To address these issues, this paper proposes an improved object detection framework based on YOLOv11, enhanced by two novel modules: the MixStructureBlock and Efficient Multi-scale Attention (EMA). The MixStructureBlock integrates multi-scale dilated convolutions and hybrid attention mechanisms to strengthen the feature extraction capacity of the backbone, while EMA refines feature maps in the detection head through spatial-channel fusion and grouped attention. The model is trained and evaluated on the TrashCan-Instance and TrashCan-Material datasets. Experimental results demonstrate that the proposed method achieves superior detection accuracy, reaching mAP@0.5 of 81.54% and 82.75% on the two datasets respectively, outperforming baseline detectors such as YOLOv11, YOLOv8, YOLOTrashCan and further surpassing recent underwater-specific SOTA models including TC-YOLO and YOLOv8-MU, achieving the competitive and superior performance among the compared methods. Ablation studies further verify the individual contributions of each module. The proposed architecture achieves an excellent balance between accuracy and efficiency, making it suitable for real-time marine debris monitoring applications. Our code is available at: https://github.com/DavidJing777/yolo-marine.

**Keywords** Object Detection, YOLOv11, Marine Debris, Deep Learning, Marine Pollution

## 1. Introduction

With the rapid industrialization of coastal regions and the continuous expansion of maritime activities, the accumulation of underwater debris has emerged as a critical global environmental issue. Marine debris adversely impacts aquatic ecosystems, threatens marine organisms through ingestion and entanglement, and disrupts the operation of underwater robotic systems vital for environmental monitoring and seabed exploration. In particular, rivers have been identified as significant pathways for transporting plastic waste into oceans, exacerbating the global debris problem.

Given the urgent need for effective monitoring, accurate and real-time marine debris detection has become essential for supporting intelligent marine surveillance systems and autonomous cleanup initiatives. However, the underwater environment presents significant challenges for visual detection due to low visibility, dynamic lighting conditions, small-scale and overlapping debris, and high background clutter[1].

Recent advances in deep learning and computer vision have paved the way for more effective marine debris detection systems[2]. Early efforts have explored lightweight embedded algorithms for complex underwater environments [3], robotic vision integration for in-situ marine litter detection, and aerial image analysis for floating macro-litter monitoring [4]. Techniques from adjacent fields, such as space debris detection, further illustrate the feasibility of deep learning-based methods in noisy, sparse object detection tasks [5]. Recent studies have adapted YOLO-family detectors specifically for underwater imagery, addressing turbidity, color cast, and the prevalence of small, occluded objects. For example, TC-YOLO integrates attention mechanisms with image enhancement for improved detection in turbid waters[6]. YOLOv8-MU adopts a large-kernel block and multi-branch reparameterization to broaden effective receptive fields without large computational overhead[7], while Underwater-YOLO and several lightweight variants (e.g., LFN-YOLO) employ dilated/deformable convolutions and occlusion-aware attention to boost small-object recall in underwater scenes[8,9].

The YOLO (You Only Look Once) series [10] has played a significant role in real-time object detection, offering high accuracy with rapid inference. Building upon YOLOv4, the YOLOTrashCan network [11] introduced feature fusion and backbone enhancements specifically for underwater debris detection, achieving an mAP@0.5 of 65.01% on the TrashCan-Instance dataset, outperforming SSD, EfficientDet [12], YOLOv3, and YOLOv4. However, effectively capturing small-scale, occluded, and semantically similar debris remains challenging.

More recently, YOLOv11 has been proposed with several architectural improvements, such as enhanced multi-scale feature extraction and lightweight attention modules. Its successful applications in vehicle detection [13] and medical imaging tasks such as bone fracture classification [14] demonstrate its adaptability across domains. Furthermore, recent years have seen increasing efforts to adapt YOLO-family detectors to underwater environments, focusing on issues such as turbidity, color distortion, low contrast, and small-object visibility. Several recent underwater-oriented YOLO variants introduce color correction, large-kernel feature extraction, lightweight attention, or multi-branch structures to enhance robustness in challenging underwater scenes[6,7,15]. These works highlight two important trends: (1) the need for rich multi-scale representation to capture small and occluded debris and (2) the importance of attention-based contextual refinement to suppress underwater background noise. Inspired by these advancements and the recent success of attention-augmented deep networks, this work proposes an improved YOLOv11-based framework that incorporates a novel MixStructureBlock and an Efficient Multi-scale Attention (EMA) module. These innovations aim to enhance the feature extraction capability and localization accuracy, especially in challenging underwater environments.

## 2. Related works

The task of marine debris detection has been extensively studied with the aid of deep learning technologies. Deng et al. [3] proposed an embeddable detection algorithm designed for underwater complexity, emphasizing low computational costs. Fulton et al. [16] integrated deep detection models into robotic platforms for in-situ identification of marine litter, while Garcia-Garin et al. [4] developed a deep learning system for floating debris detection from aerial imagery, linked with a real-time web application. Complementary studies, such as Massimi et al. [5] in the context of space debris detection, reinforce the potential of deep learning for handling sparse, noisy detection problems in dynamic environments.

Attention mechanisms have emerged as crucial components in boosting detection performance under challenging conditions. Methods like EAPT introduced pyramid-based efficient attention mechanisms for better multi-scale feature aggregation. FSAD-Net [17] demonstrated that feedback spatial attention can significantly improve robustness in visibility-impaired scenarios such as dehazing and underwater vision. Moreover, works like BaGFN and Multimodal Cascaded CNNs highlighted the benefits of high-order feature interactions and multimodal learning in improving object classification and localization accuracy.

For small object detection and fine-grained classification tasks, Sthy-Net [18] proposed a dense-branched module design to enhance feature fusion, which is particularly beneficial for detecting small debris elements underwater. Furthermore, enhanced underwater detection frameworks [19] combining attention mechanisms with dilated large-kernel convolutions have proven effective in dealing with the specific challenges of underwater clutter and scale variation.

Building upon the YOLO family, YOLOTrashCan [11] modified YOLOv4 by introducing attention mechanisms and feature pyramid enhancements, achieving notable improvements in marine debris detection. YOLOv11 further advanced this line of work with architectural refinements aimed at multi-scale feature extraction and efficient context modeling. Applications across transportation and medical imaging attest to its robustness and adaptability.

In recent years, several YOLO-based architectures have been specifically designed for underwater detection tasks. TC-YOLO integrates attention modules with color-corrective preprocessing to reduce the impact of turbidity and color cast in underwater images[6]. AWF-YOLO employs an adaptive weighted feature pyramid network to improve multi-scale fusion and enhance the detection of small marine targets[20]. Similarly, UGC-YOLO introduces a global context block to better capture long-range dependencies in visually complex underwater backgrounds. More recently, improved YOLOv8 variants such as YOLOv8-MU and SPSM-YOLOv8 incorporate large-kernel convolutions and lightweight reparameterized modules to achieve stronger receptive field aggregation and higher efficiency[7,21]. These underwater-oriented detection networks consistently demonstrate that enhanced multi-scale representation and attention mechanisms are crucial for improving robustness and accuracy in underwater environments.

Motivated by these prior efforts, our work proposes a refined YOLOv11-based detector, integrating MixStructureBlock and EMA modules to further enhance detection performance for marine debris, focusing on small-scale, occluded, and visually complex underwater targets.

## 3. Methodology

This section describes the overall architecture of the proposed method, which is built upon the YOLOv11 object detection framework. In order to enhance the model's capacity for underwater marine debris detection, we introduce two novel structural components: The MixStructureBlock, embedded into the backbone, and the EMA attention module, integrated into the detection head. These improvements are designed to address common challenges in underwater detection tasks such as small object localization, occlusion, and background interference.

The overall architecture of our model is illustrated in Figure 1, and consists of three main components:

- A feature extraction backbone with multi-scale fusion and attention enhancement;
- A modified neck structure for hierarchical feature aggregation;
- An improved detection head with Efficient Multi-scale Attention (EMA) for refined prediction.

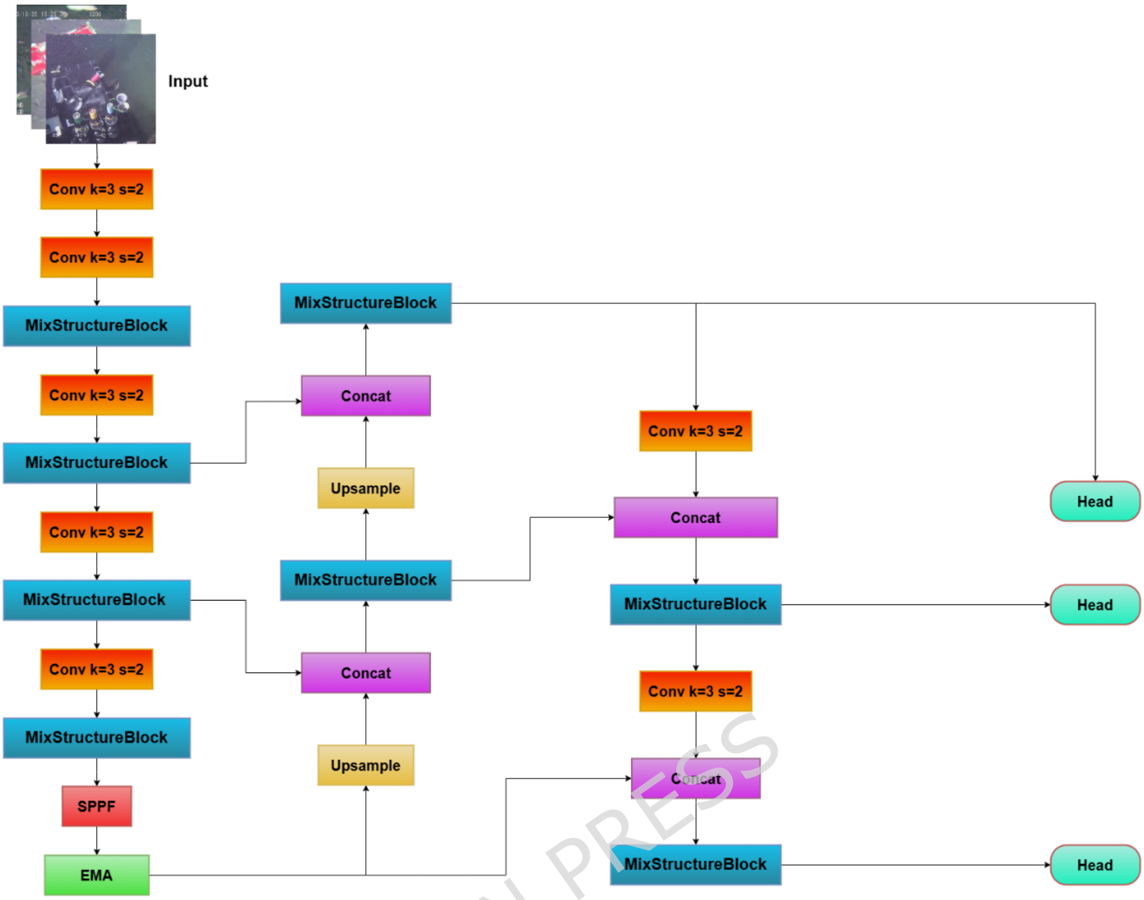



**Fig. 1** Structure of our proposed model. The MixStructureBlock module replaces the C3k2 module in yolov11 and the EMA module replaces the C2PSA module.

### *3.1. MixStructureBlock*

The backbone of the original YOLOv11 consists of convolutional layers and residual blocks designed to extract rich visual features at multiple spatial resolutions. Although the original C3k2 module in YOLOv11 provides lightweight feature extraction, it still relies on fixed-size convolutions and limited receptive field patterns. This constraint makes it difficult to capture long-range contextual cues and multi-scale spatial variations that are common in underwater scenes, where debris objects often appear small, occluded, or blended into noisy backgrounds.

To address these challenges, inspired by YoloTrashCan's introduction of Do-Con[22], we introduce the MixStructureBlock[23], as is shown in Figure 2, a hybrid block that combines multi-branch dilated convolutions and compound attention mechanisms into the backbone. The use of dilation enables larger receptive fields without increasing computational cost, while the parallel branches capture diverse spatial structures ranging from fine textures to global contours. Furthermore, the integrated channel–spatial attention selectively emphasizes informative features under low contrast and high background clutter. This module replaces standard C3k2 blocks in YOLOv11 and provides two core enhancements:

1. Multi-scale feature extraction

Within the block, parallel convolutional paths with different kernel sizes (e.g., 3×3, 5×5, 7×7) and dilations (e.g., dilation=3) are applied to the input features. These multiple receptive fields allow the block to capture local textures and long-range dependencies simultaneously, which is particularly beneficial for debris objects that appear in varied sizes and contexts.

2. Integrated attention mechanisms

- Channel Attention (CA): Uses global average pooling followed by a sigmoid-activated MLP to reweight feature channels, emphasizing informative object categories;
- Pixel Attention (PA): Highlights salient spatial locations using 1×1 convolutions and activation functions;
- Simple Pixel Attention (SPA): Enhances spatial focus by combining grouped convolutions with spatial gating.

These branches are followed by a channel concatenation and a 1×1 MLP-based fusion layer that compresses the multi-branch output into the original channel dimension. Residual connections are used throughout to stabilize training and maintain gradient flow.

These properties make MixStructureBlock a more suitable replacement for C3k2 when handling underwater debris, where multi-scale textures and ambiguous object boundaries require richer contextual modeling. This design enhances the model's capability to detect marine debris across various scales, improves robustness to noise and clutter, and significantly boosts the backbone's representation power without incurring excessive computational cost. This design is inspired by recent large-kernel and reparameterization studies which demonstrate that expanded receptive fields can significantly improve small-object and contextual representation (RepLKNet, DO-Conv, ODConv)[22,24,25].

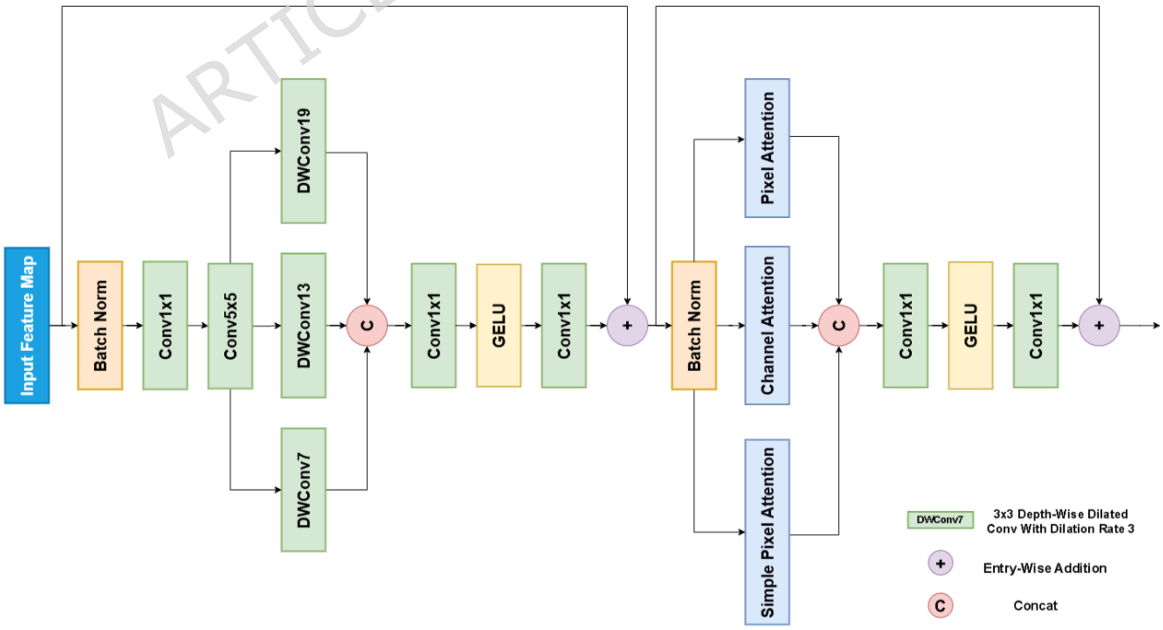



**Fig. 2** The framework of MixStructureBlock

*3.2. EMA*

In recent years, attention mechanisms represented by CBAM[26], ResNsSt , and ECA [27] have been widely adopted, as they have been proven to significantly improve the performance of deep neural networks.

To enhance the model's ability to perceive and distinguish marine debris in visually complex underwater environments, we replace the original C2PSA module in YOLOv11 with an Efficient Multi-scale Attention (EMA) module [28]. The introduce EMA is designed to improve multi-scale feature extraction and adaptive focus on salient regions, particularly benefiting the detection of small, overlapping, and low-contrast objects commonly found in underwater scenes. Many experiments have demonstrated that the introduction of EMA into YOLO improves detection performance. In [28], EMA was integrated into YOLOv5s and YOLOv5x and the mAP of the model was significantly improved. In [29], the introduction of ODConv[30] and EMA into yolov8 for Road Pattern Recognition detection achieved an accuracy of 93.2%. Figure 3 illustrates the framework of the EMA module.

The original C2PSA module focuses primarily on pixel-wise spatial attention, which limits its ability to model long-range contextual dependencies and multi-level semantic interactions. Underwater debris images commonly exhibit low contrast, turbidity, and overlapping targets, where purely spatial attention cannot adequately separate foreground from background.

EMA offers a more comprehensive mechanism by jointly modeling global channel descriptors and multi-scale spatial context through grouped pooling and cross-dimensional feature fusion. This design enhances sensitivity to small objects and improves robustness against visual degradation, making EMA a more effective alternative to C2PSA in underwater scenarios.

The EMA module fuses both spatial and channel-wise attention mechanisms in a lightweight and computation-efficient manner. Structurally, it is composed of the following key components:

- Multi-scale spatial context modeling: The input features are passed through parallel 3×3 convolutions and adaptive average pooling to capture both local and global spatial dependencies;
- Group normalization and weighting: Group-wise normalization (GroupNorm) and Softmax activation are applied to regulate intra-group feature distribution and adaptively reweight important feature channels;
- Cross-dimensional attention fusion: Using matrix multiplication between pooled spatial features and channel descriptors, EMA captures high-order dependencies across dimensions. A Sigmoid activation is applied to generate a final attention map for dynamic feature modulation.

As shown in the modified network architecture, the EMA module is embedded at the end of the Neck, replacing the original C2PSA in YOLOv11. It functions as a lightweight attention unit that recalibrates feature responses across both space and channels, effectively suppressing background noise while enhancing discriminative features. The EMA draws on recent efficient attention and dynamic convolution works that emphasize compact multi-scale cross-dimensional feature fusion (ODConv, ECA, MobileViT), justifying our use of grouped pooling and cross-dimensional fusion for lightweight but effective attention.

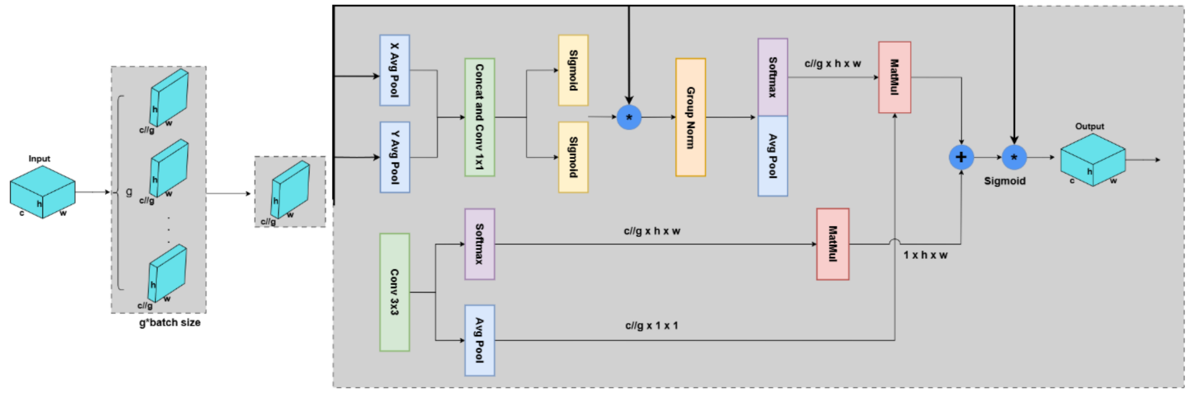



**Fig. 3** The framework of EMA

### *3.3 Model Architecture*

The proposed model is an enhanced version of YOLOv11, designed to improve detection performance under challenging underwater conditions. It consists of a backbone for feature extraction, a feature fusion neck, and a detection head, as illustrated in Figure 1.

The network begins with a series of convolutional layers with stride 2 that downsample the input image and increase the feature dimension. At each stage, the feature maps are processed by the MixStructureBlock, which is designed to capture richer contextual information. This block combines multi-branch convolutions of varying receptive fields and integrates both local and global dependencies, allowing the model to better handle objects of different shapes and scales.

Following the backbone, the deepest feature map is processed by a Spatial Pyramid Pooling-Fast (SPPF) module to enhance the receptive field. Subsequently, the output is refined by the Efficient Multi-scale Attention (EMA) module, which replaces the original C2PSA module from YOLOv11. EMA aggregates spatial and channel information through grouped pooling and efficient attention fusion, improving the model's ability to focus on informative regions while suppressing background noise.

The neck adopts a PANet-style top-down pathway, where high-resolution features are generated by upsampling and concatenating deeper features with shallower ones. After fusion, additional MixStructureBlock modules are applied to further refine the semantic representation. This is followed by a bottom-up pathway, where feature maps are downsampled and concatenated again to enrich contextual information across multiple levels.

Finally, three different feature levels are fed into parallel detection heads. Each detection head predicts object bounding boxes, class probabilities, and confidence scores. The outputs from all detection heads are combined, and non-maximum suppression (NMS) is applied to obtain the final detection results.

Overall, the controlled ablation experiments demonstrate that each component contributes independently to performance improvement, providing clear justification for replacing C3k2 with MixStructureBlock and C2PSA with EMA in the proposed architecture.

## 4. Experimental Results and Analysis

*4.1. Dataset*

To effectively train and evaluate our proposed underwater debris detection model, we utilize the TrashCan dataset, a large-scale benchmark specifically designed for marine vision tasks. The dataset was constructed from the J-EDI deep-sea image electronic library, maintained by the Japan Agency of Marine Earth Science and Technology (JAMSTEC). It comprises over 1000 underwater videos captured across various oceanic regions around Japan, under diverse lighting and environmental conditions.

From these videos, a total of 7212 RGB images were extracted and annotated. The dataset reflects real-world underwater complexity, including various forms of marine debris, biological organisms (e.g., fish, coral), underwater equipment, and cluttered backgrounds. To support different detection scenarios, TrashCan is provided in two labeling formats:

- TrashCan-Instance: Contains 22 object categories, including specific debris types such as trash bag, trash cup, trash branch, as well as bio (flora/fauna), rov (remotely operated vehicles), and unknown classes;
- TrashCan-Material: Contains 16 categories, classified based on material type (e.g., plastic, metal, fabric).

In our experiments, we adopt the TrashCan-Instance version due to its finer object-level granularity, which is well-suited for evaluating multi-class object detection performance. The dataset is split into a training set of 6065 images and a validation set of 1147 images. Across the two sets, a total of 12,128 labeled object instances are provided, covering a broad range of object sizes, shapes, and appearances.



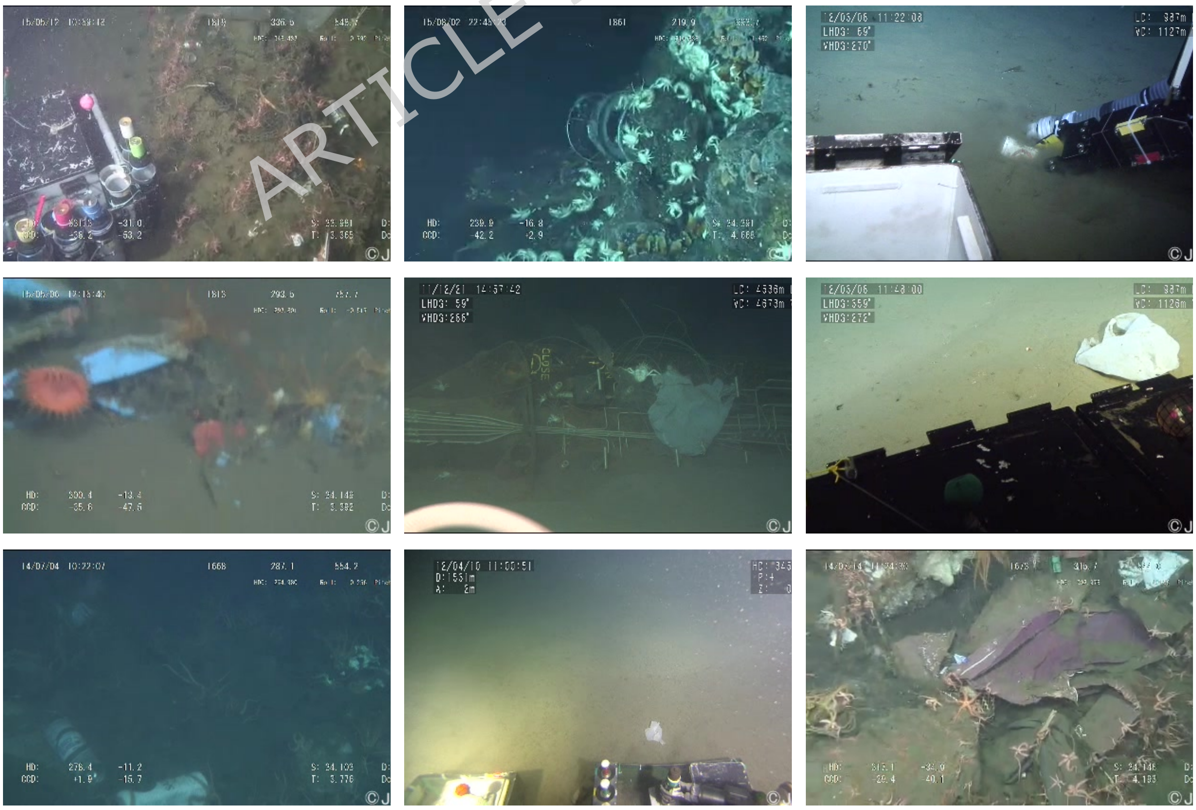


**Fig. 3** Images of TrashCan dataset

### 4.2. Evaluation Metrics

To comprehensively evaluate the performance of the proposed object detection model, we employ a set of standard metrics widely used in the object detection domain, particularly in the context of the COCO and PASCAL VOC evaluation protocols. The key evaluation criteria include Precision, Recall, F1-score, and mean Average Precision (mAP).

Precision measures the proportion of correctly predicted positive samples among all predicted positives, indicating the detector's ability to avoid false positives. It is defined as:

$$\text{Precision} = \frac{TP}{TP+FP} \tag{1}$$

Recall measures the proportion of correctly predicted positive samples among all ground-truth positives, reflecting the detector's ability to capture relevant objects. It is defined as:

$$\text{Recall} = \frac{TP}{TP+FN} \tag{2}$$

F1-score is the harmonic mean of Precision and Recall, providing a balanced indicator of detection performance. It is defined as:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

Mean Average Precision (mAP@0.5) is the average of AP scores across all object categories at IoU = 0.5. We also report mAP@0.5:0.95, which averages AP across IoU thresholds from 0.5 to 0.95 with a step size of 0.05, providing a more stringent and comprehensive evaluation.

$$\text{AP} = \int_0^1 p(r)\,dr \tag{4}$$

Where p(r) is a function of Precision about Recall.

$$\text{mAP} = \frac{1}{N}\sum_{i=1}^{N} \text{AP}_i \tag{5}$$

Model size (MB) and parameter count are reported to assess the computational efficiency and deployment potential of the proposed architecture.

### 4.3. Experimental Environment and Training

All experiments were conducted on a device equipped with Windows 10, an Intel Core i5-12400F CPU @ 2.50 GHz, 16 GB RAM, and an NVIDIA GeForce RTX 3060 GPU with 12 GB memory. The training was implemented using PyTorch 2.1.0+cu121, with CUDA 12.1 and the Ultralytics YOLO framework.

For all training sessions, the input images were resized to a resolution of 416×416 pixels. The models were trained for 180 epochs using a batch size of 128, and cosine learning rate scheduling was applied to stabilize the convergence process and prevent overfitting. All other parameters were kept as default unless otherwise specified.

### 4.4. Results

#### 4.4.1. Experiment on Trash-Instance

Although several recently proposed underwater-specific YOLO variants such as AWF-YOLO, SPSM-YOLOv8, and LFN-YOLO demonstrate promising performance, many of them lack publicly available codebases, pretrained weights, or repeatable training pipelines. To ensure that all comparisons are fair and reproducible, the experimental evaluation in this work includes only models with openly accessible implementations and retrainable configurations on the TrashCan-Instance dataset. Accordingly, the benchmark models consist of two reproducible underwater state-of-the-art detectors (TC-YOLO and YOLOv8-MU), one marine-debris-specialized model (YOLOTrashCan), two representative general-purpose detectors (YOLOv8 and YOLOv11), and our proposed model. Other recent underwater YOLO variants are discussed qualitatively in the Related Works section, where their design principles and architectural motivations are analyzed in relation to our approach.

**Table 1.** Performance of different models on TrashCan-Instance.

| | **TC-YOLO** | **YOLOv8-MU** | **YOLOTrashCan[11]** | **YOLOv8** | **YOLOv11** | **Ours** |
|---|---|---|---|---|---|---|
| Input | 640 x 640 | 640 x 640 | 416 x 416 | 416 x 416 | 416 x 416 | 416 x 416 |
| Type | Underwater SOTA | Underwater SOTA | Marine debris | General | General | Proposed |
| Backbone | Transformer_CA | UniRepLKNet_C2fSTR | ECA_DO_Conv_CSPDarknet53 | C2f | C3k2_C2PSA | MixStructureBlock_EMA |
| Size(MB) | 6.34 | 6.95 | 214 | 5.95 | 7.50 | 5.20 |
| mAP@0.5 | 75.40% | 76.19% | 65.01% | 67.83% | 68.06% | 81.54% |

Table 1 presents a comprehensive comparison of model performance on the TrashCan-Instance dataset. The results show that our method achieves the highest overall accuracy, reaching an mAP@0.5 of 81.54%. This significantly surpasses the two underwater SOTA models TC-YOLO and YOLOv8-MU, both of which exhibit strong baseline performance but remain several points below our proposed method. Meanwhile, YOLOTrashCan, although specifically designed for marine debris detection, performs noticeably lower, highlighting the limitations of its heavier backbone and older architectural design. Compared with general-purpose detectors such as YOLOv8 and YOLOv11, our model also demonstrates clear improvements, confirming that the introduced MixStructureBlock and EMA attention mechanism effectively enhance feature representation and object localization capability in complex underwater environments.

Despite delivering the best detection accuracy, our model maintains a lightweight size of only 5.20 MB. This is smaller than YOLOv11 and YOLOv8, and dramatically more compact compared with the substantially larger YOLOTrashCan model. Such compactness, combined with improved accuracy, demonstrates the efficiency of our architectural refinements and underscores the practicality of deploying the proposed model in real-time embedded underwater systems, where computational resources are often limited.

To intuitively demonstrate the detection performance improvement brought by our proposed model, Figure 5 presents visual comparisons on the TrashCan-Instance dataset across three groups: (a) the ground-truth labels, (b) predictions from the original YOLOv11, and (c) predictions from our improved model. Each row corresponds to one scene, covering various challenging underwater conditions such as occlusion, class ambiguity, and complex background.

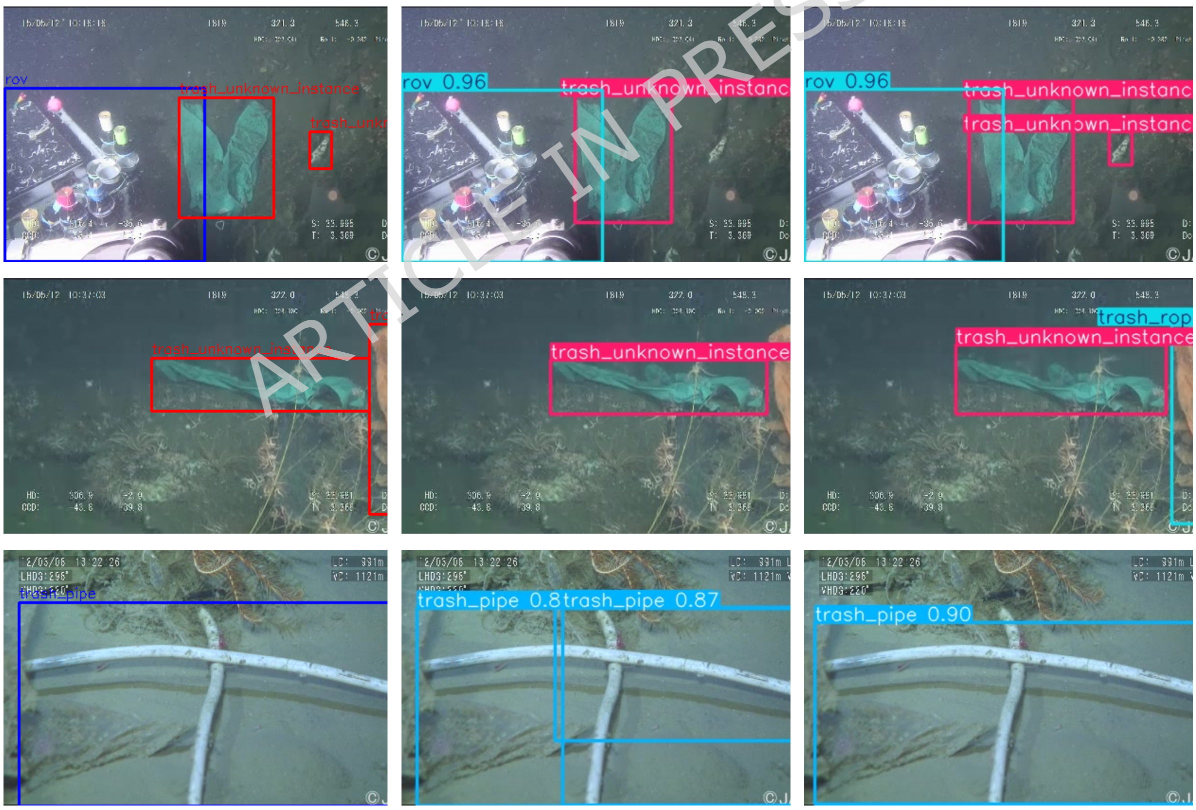
In the first row, both YOLOv11 and our model are able to detect multiple instances of trash_unknown_instance. However, our model outputs higher confidence scores and more accurate bounding box localization, especially for the smaller object, which YOLOv11 tends to miss or predict with lower confidence.

In the second row, our model suppresses redundant bounding boxes and correctly differentiates adjacent or overlapping objects. While YOLOv11 identifies the main object, its prediction suffers from imprecise boundaries and redundant box outputs.

In the third row, for the *trash_pipe* category, our model produces a more confident and cleaner prediction with better alignment to the object contour, compared to the slightly fragmented detection by YOLOv11.

These qualitative results show that our proposed improvements—namely the MixStructureBlock and the EMA attention mechanism—help the model focus on informative features while maintaining robustness in cluttered underwater scenes. Compared to the baseline YOLOv11, our model yields more accurate detections with fewer false positives and stronger consistency in multi-object scenarios.






(**a**) (**b**) (**c**)

**Fig. 5** Detection of YOLOv11 and our model on TrashCan-Instance: (a) True label; (b) YOLOv11; (c) Our model.

*4.4.2. Ablation Experiment*

To investigate the effectiveness of the proposed architectural improvements, we conducted a series of ablation experiments on the TrashCan-Instance dataset.

Four variants were compared: the baseline YOLOv11 model, YOLOv11+EMA, YOLOv11+MixStructureBlock, and the full model with both components integrated. Table 2 summarizes the performance across four variants in terms of precision, recall, F1-score, and mAP@0.5.

The baseline YOLOv11 model achieves a precision of 82.08%, recall of 64.06%, and an mAP@0.5 of 68.06%. By replacing the original C2PSA module with the proposed EMA, the model (YOLOv11 + EMA) improves recall to 76.31% and F1-score to 0.79, resulting in an mAP@0.5 of 78.93%. This indicates that EMA effectively enhances contextual feature learning and better captures multi-scale dependencies, particularly beneficial for cluttered underwater environments.

Introducing the MixStructureBlock alone (YOLOv11 + MixStructure) significantly increases precision to 87.11%, the highest among all variants, and boosts mAP@0.5 to 79.80%. This confirms that MixStructureBlock improves the detector's capacity for spatial feature extraction and reduces false positives, especially for small or overlapping objects.

When both modules are combined in our final model (Ours), performance is further enhanced. The F1-score reaches 0.84, and mAP@0.5 improves to 81.54%, demonstrating the complementary effect of the two modules. The balanced gains in both precision and recall suggest improved detection robustness and classification accuracy.

**Table 2.** Results of ablation experiment. Ablation experiments demonstrating the effect of different modules on detection results.

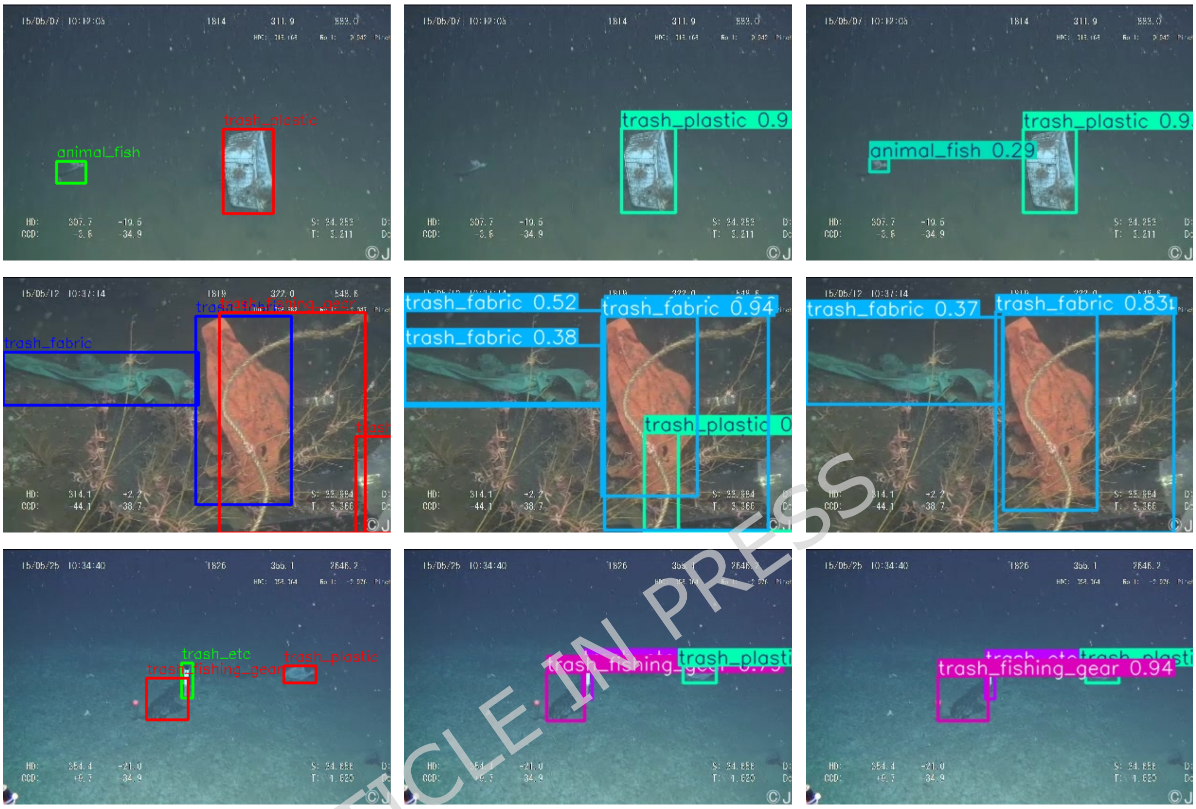
| Model | Precision(%) | Recall(%) | F1-score | mAP@0.5(%) |
|---|---|---|---|---|
| YOLOv11 | 82.08 | 64.06 | 0.72 | 68.06 |
| YOLOv11+EMA | 81.12 | 76.31 | 0.79 | 78.93 |
| YOLOv11+MixStructureBlock | 87.11 | 75.05 | 0.81 | 79.80 |
| Ours | 82.05 | 86.75 | 0.84 | 81.54 |

The controlled experiments clearly isolate the effect of each proposed module. When only EMA is added, the model achieves a higher recall (+12.25%) and F1-score, indicating that EMA improves sensitivity to small and low-contrast objects by enhancing global contextual reasoning. By contrast, adding only the MixStructureBlock significantly increases precision (+5.03%), demonstrating that the hybrid dilated convolution structure reduces false positives by capturing richer spatial cues and improving feature discrimination in cluttered underwater scenes.

When the two modules are combined, their complementary strengths result in simultaneous gains in both precision and recall, leading to the highest overall mAP@0.5 of 81.54%. This confirms that MixStructureBlock and EMA address different weaknesses of the baseline YOLOv11: the former improves spatial structure representation, while the latter enhances multi-scale attention and global context aggregation. Their joint effect justifies the architectural modification adopted in the final model.

*4.4.3. Experiment on TrashCan-Material*

To further evaluate the effectiveness of the proposed model, we conducted experiments on the TrashCan-Material dataset and compared our results with several baseline models, including TC-YOLO, YOLOv8-MU, YOLOTrashCan, YOLOv8, and YOLOv11. As shown in Table 3, our model achieves the highest detection accuracy with an mAP@0.5 of 82.75%, surpassing YOLOv11 (70.06%) and YOLOTrashCan (58.66%). Figure 6 shows qualitative detection results on the TrashCan-Material dataset, comparing ground-truth annotations, YOLOv11 predictions, and the results of our proposed model under complex underwater conditions.




(**a**) (**b**) (**c**)

**Fig. 6** Detection of YOLOv11 and our model on TrashCan-Material: (a) True label; (b) YOLOv11; (c) Our model



Despite being significantly lighter in size (5.20MB), our model maintains strong performance, highlighting the efficiency of the proposed architecture. This improvement is mainly attributed to the integration of the MixStructureBlock and EMA modules, which effectively enhance multi-scale feature extraction and attention-based feature refinement in complex underwater environments.

**Table 3.** Performance of different models on TrashCan-Material.

| | **TC-YOLO** | **YOLOv8-MU** | **YOLOTrashCan[11]** | **YOLOv8** | **YOLOv11** | **Ours** |
|---|---|---|---|---|---|---|
| Input | 640 x 640 | 640 x 640 | 416 x 416 | 416 x 416 | 416 x 416 | 416 x 416 |
| Type | Underwater SOTA | Underwater SOTA | Marine debris | General | General | Proposed |
| Backbone | Transformer_CA | UniRepLKNet_C2fSTR | ECA_DO_Conv_CSPDarknet53 | C2f | C3k2_C2PSA | MixStructure Block_EMA |
| Size(MB) | 6.34 | 6.95 | 214 | 5.95 | 7.50 | 5.20 |
| mAP@0.5 | 75.40% | 76.19% | 58.66% | 69.73% | 70.06% | 82.75% |

## 5. Conclusion

In this study, we proposed an improved YOLOv11-based architecture for underwater marine debris detection. To address the challenges posed by complex underwater environments—such as object occlusion, scale variation, and background noise—we introduced two key modules: the MixStructureBlock, which enhances multi-scale feature representation through hybrid convolutional structures, and the EMA module, which adaptively focuses on informative spatial and channel-wise features.

Extensive experiments on the TrashCan-Instance and TrashCan-Material datasets demonstrated the effectiveness of the proposed model. Compared to the baseline YOLOv11 and other state-of-the-art underwater-specific models, our method achieved superior performance, with the highest mAP@0.5 of 81.54% on TrashCan-Instance and 82.75% on TrashCan-Material. Meanwhile, our model size is lighter compared to previous studies at only 5.2MB. Ablation studies further confirmed the individual and combined contributions of EMA and MixStructureBlock to detection accuracy and robustness.

## References


1. Chen, L., et al., *Underwater optical object detection in the era of artificial intelligence: current, challenge, and future.* ACM Computing Surveys, 2025. **58**(3): p. 1-34.
2. Elmezain, M., et al., *Advancing underwater vision: a survey of deep learning models for underwater object recognition and tracking.* IEEE Access, 2025.
3. Deng, H., et al., *An embeddable algorithm for automatic garbage detection based on complex marine environment.* Sensors, 2021. **21**(19): p. 6391. http://dx.doi.org/10.3390/s21196391
4. Garcia-Garin, O., et al., *Automatic detection and quantification of floating marine macro-litter in aerial images: Introducing a novel deep learning approach connected to a web application in R.* Environmental Pollution, 2021. **273**: p. 116490. http://dx.doi.org/10.1016/j.envpol.2021.116490
5. Massimi, F., et al., *Deep learning‐based space debris detection for space situational awareness: A feasibility study applied to the radar processing.* IET Radar, Sonar & Navigation, 2024. **18**(4): p. 635-648. http://dx.doi.org/10.1049/rsn2.12547
6. Liu, K., L. Peng, and S. Tang, *Underwater object detection using TC-YOLO with attention mechanisms.* Sensors, 2023. **23**(5): p. 2567.
7. Jiang, X., et al., *YOLOv8-MU: an improved YOLOv8 underwater detector based on a large kernel block and a multi-branch reparameterization module.* Sensors, 2024. **24**(9): p. 2905.
8. Li, Z., et al., *Underwater-Yolo: Underwater Object Detection Network with Dilated Deformable Convolutions and Dual-Branch Occlusion Attention Mechanism.* Journal of Marine Science and Engineering, 2024. **12**(12): p. 2291.
9. Liu, M., et al., *LFN-YOLO: precision underwater small object detection via a lightweight reparameterized approach.* Frontiers in Marine Science, 2025. **11**: p. 1513740.
10. Redmon, J., et al. *You only look once: Unified, real-time object detection*. in *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2016.

11. Zhou, W., et al., *YOLOTrashCan: a deep learning marine debris detection network.* IEEE Transactions on Instrumentation and Measurement, 2022. **72**: p. 1-12. http://dx.doi.org/10.1109/TIM.2022.3225044
12. Tan, M., R. Pang, and Q.V. Le. *Efficientdet: Scalable and efficient object detection*. in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2020.
13. Bakirci, M., et al. *Multi-Class Vehicle Detection and Classification with YOLO11 on UAV-Captured Aerial Imagery*. in *2024 IEEE 7th International Conference on Actual Problems of Unmanned Aerial Vehicles Development (APUAVD)*. 2024. IEEE.
14. Wei, W., et al., *YOLOv11-based multi-task learning for enhanced bone fracture detection and classification in X-ray images.* Journal of Radiation Research and Applied Sciences, 2025. **18**(1): p. 101309. http://dx.doi.org/10.1016/j.jrras.2025.101309
15. Zhang, F., et al., *Underwater object detection algorithm based on an improved YOLOv8.* Journal of Marine Science and Engineering, 2024. **12**(11): p. 1991.
16. Fulton, M., et al. *Robotic detection of marine litter using deep visual detection models*. in *2019 international conference on robotics and automation (ICRA)*. 2019. IEEE.
17. Zhou, Y., et al., *FSAD-Net: feedback spatial attention dehazing network.* IEEE transactions on neural networks and learning systems, 2022. **34**(10): p. 7719-7733. http://dx.doi.org/10.1109/TNNLS.2022.3146004
18. Al-Jebrni, A.H., et al., *Sthy-net: a feature fusion-enhanced dense-branched modules network for small thyroid nodule classification from ultrasound images.* The Visual Computer, 2023. **39**(8): p. 3675-3689. http://dx.doi.org/10.1007/s00371-023-02984
19. Ouyang, J. and Y. Li, *Enhanced underwater object detection via attention mechanism and dilated large-kernel networks.* The Visual Computer, 2025. http://dx.doi.org/10.1007/s00371-025-03870-4
20. Guo, Q., et al., *AWF-YOLO: enhanced underwater object detection with adaptive weighted feature pyramid network*. 2023: OAE Publishing Inc.
21. Cheng, S., et al., *Attention-Based Lightweight YOLOv8 Underwater Target Recognition Algorithm.* Sensors, 2024. **24**(23): p. 7640.
22. Cao, J., et al., *Do-conv: Depthwise over-parameterized convolutional layer.* IEEE Transactions on Image Processing, 2022. **31**: p. 3726-3736. http://dx.doi.org/10.1109/TIP.2022.3175432
23. Lu, L., et al. *MixDehazeNet: Mix structure block for image dehazing network*. in *2024 International Joint Conference on Neural Networks (IJCNN)*. 2024. IEEE.
24. Ding, X., et al. *Scaling up your kernels to 31x31: Revisiting large kernel design in cnns*. in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2022.
25. Li, C., A. Zhou, and A. Yao, *Omni-dimensional dynamic convolution.* arXiv preprint arXiv:2209.07947, 2022.
26. Woo, S., et al. *CBAM: Convolutional block attention module*. in *Proceedings of the European conference on computer vision (ECCV)*. 2018.
27. Wang, Q., et al. *ECA-Net: Efficient channel attention for deep convolutional neural networks*. in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2020.
28. Ouyang, D., et al. *Efficient multi-scale attention module with cross-spatial learning*. in *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2023. IEEE.

29. Zhang, X. and Y. Yang, *Research on Road Pattern Recognition of a Vision-Guided Robot Based on Improved-YOLOv8.* Applied Sciences, 2024. **14**(11): p. 4424. http://dx.doi.org/10.3390/app14114424
30. Zhao, Z. and M. Dong. *Channel-spatial dynamic convolution: an exquisite omni-dimensional dynamic convolution*. in *2023 8th international conference on intelligent computing and signal processing (ICSP)*. 2023. IEEE.

**Author contributions**

Jing Yuanwei conceived the study, implemented the programming, conducted model training, and drafted the original manuscript. Ding Yijiang was responsible for data collection, testing, validation, and manuscript editing. Wang Xuemei contributed to data collection, testing, validation, and information retrieval. Anis Salwa Mohd Khairuddin supervised the project and critically reviewed the manuscript. All authors reviewed the manuscript, approved the final version, and agree to be accountable for all aspects of the work, ensuring that questions related to the accuracy or integrity of any part are appropriately addressed.

**Data availability**

The datasets used in this study are publicly available from the TrashCan benchmark provided by the Japan Agency for Marine-Earth Science and Technology (JAMSTEC) at https://www.godac.jamstec.go.jp/catalog/dsdebris/e/index.html. Additional information is available from the corresponding author upon reasonable request.

**Declarations**

The authors declare no competing interests.

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